18-7-24

Waves Lec 15

Power Absorbed by a driven osc.

Undamped osc.

$$\frac{d^2x}{dt^2} + \omega_0^2 x = \frac{F_0}{m}\cos\omega t$$

$$x = A\cos(\omega t + \alpha)$$

$$A = \begin{cases} \dfrac{F_0}{m(\omega_0^2-\omega^2)}, & \omega < \omega_0 \\ \dfrac{F_0}{m(\omega^2-\omega_0^2)}, & \omega > \omega_0 \end{cases}$$

$$\alpha = \begin{cases} 0, & \omega < \omega_0 \\ \pi, & \omega > \omega_0 \end{cases}$$

$$\boxed{x = C\cos\omega t}$$

$$(\omega_0^2 - \omega^2)C = \frac{F_0}{m}$$

$$\boxed{C = \frac{F_0}{m(\omega_0^2-\omega^2)}}$$

$$v = \dot{x} = -\omega C\sin\omega t$$

$$\dot{W} = P = F\frac{dx}{dt} = Fv$$

$$= F_0\cos\omega t\,(-\omega C)\sin\omega t$$

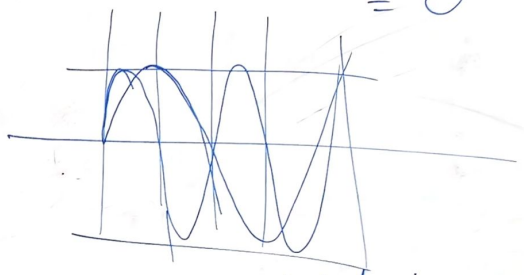
$$= -\omega C F_0 \sin\omega t\cos\omega t$$

$$= -\frac{C}{2}\omega F_0 \sin 2\omega t$$

$$\bar{P} = \frac{1}{\tau}\int_0^T P\,dt, \quad T = \frac{2\pi}{\omega}$$

$$= -\frac{C\omega F_0}{2}\frac{1}{2\pi/\omega}\int_0^{2\pi/\omega} \sin 2\omega t\,dt$$

$$= 0$$



Elastic osc without damping or friction does not dissipate absorb energy on the avg

, $T = \frac{2\pi}{\omega}$

ω

$\sin 2\omega t \, dt$

$= 0$

damping or
ssipati
on the avg

Damped and Driven OPC

$$\ddot{x} + \gamma \dot{x} + \omega_0^2 x = F_0 \cos \omega t$$

$$\boxed{x = A \cos(\omega t - \delta)}$$

$$A(\omega) = \frac{f_0}{\left[(\omega^2 - \omega_0^2)^2 + \gamma^2 \omega^2\right]^{1/2}} \quad \left| \frac{2\pi^2}{T^2} \right.$$

$$z = A e^{i(\omega t - \delta)} \qquad RHS = F_0 e^{i\omega t}$$

$$(\omega_0^2 - \omega^2) A + i\gamma\omega A = f_0 [\cos\delta + i \sin\delta]$$

$$(\omega_0^2 - \omega^2) A = f_0 \cos\delta$$

$$\gamma\omega A = f_0 \sin\delta = \frac{F_0}{m} \sin\delta$$

$$F_0 \sin\delta = m\omega\gamma A$$

$$W = P = F v$$

$$v = \dot{x} = -\omega A \sin(\omega t - \delta)$$

$$= -\underbrace{v_0(\omega)}_{\omega A} \sin(\omega t - \delta)$$

ωA = Vel. amplitude $= v_0(\omega)$

$$P = -F_0 v_0 \cos\omega t \sin(\omega t - \delta)$$

$$= -F_0 v_0 [\cos\omega t \sin\omega t \cos\delta - \cos\omega t \cos\omega t \sin\delta]$$

$$\bar{P} = F_0 v_0 \left[\underbrace{\overline{\cos^2 \omega t}}_{1/2} \sin\delta - \underbrace{\overline{\tfrac{1}{2}\sin 2\omega t}}_{0} \cos\delta\right]$$

$$\boxed{\bar{P} = \frac{1}{2} F_0 v_0 \sin\delta} = \frac{1}{2} m\omega\gamma A v_0$$

$$= \boxed{\frac{1}{2} m\omega^2 A^2 \gamma}$$

$$\bar{P} = \frac{1}{2} m\gamma\omega^2 A^2, \qquad Q = \frac{\omega_0}{\gamma}$$

$$= \frac{m}{2} f_0^2 \gamma \frac{\omega^2}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}$$

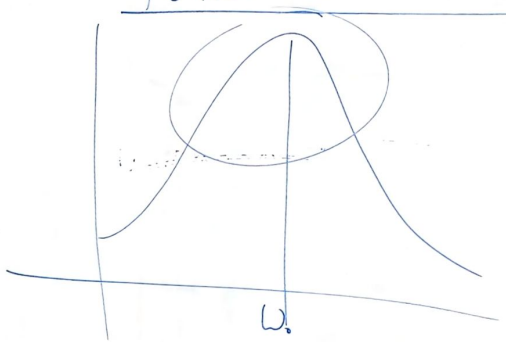
$$= \frac{m}{2} \frac{f_0^2 \gamma \omega^2}{\omega_0^2 \omega^2} \frac{1}{\left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)^2 + \frac{\gamma^2}{\omega_0^2}}$$

$$= \frac{F_0^2 \gamma}{2 m \omega_0^2} \frac{1}{\left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)^2 + \frac{1}{Q^2}}$$

18-7-26

Waves Lec 15

Power absorbed by damped-driven osc



$\omega = \omega_0 + \Delta\omega$

$= \omega_0 \left[1 + \frac{\Delta\omega}{\omega_0}\right]$

$= \omega_0 \left[1 + \eta\right]$

$\frac{\omega_0}{\omega} = \frac{1}{1+\eta} \approx 1 - \eta + \frac{\eta^2}{2} + \dots$

$\frac{\omega}{\omega_0} = 1 + \eta$

$\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0} = \left(1 - \eta + \frac{\eta^2}{2} \dots\right) - (1+\eta)$

$= -2\eta + \frac{\eta^2}{2}$

$= -2\eta \left(1 - \frac{\eta}{4} \dots\right)$

$\left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)^2 = 4\eta^2 \left(1 - \frac{\eta}{4} \dots\right)^2$

$= 4\left[\eta^2 + O(\eta^3)\right]$

$\approx 4\eta^2 = 4(\omega - \omega_0)^2/\omega_0^2$

$\bar{P}(\omega) = \frac{F_0^2 \gamma}{2m\omega_0^2} \cdot \frac{1}{4\frac{(\omega-\omega_0)^2}{\omega_0^2} + \frac{\gamma^2}{\omega_0^2}}$

$= \frac{F_0^2 \gamma}{8m} \cdot \frac{1}{(\omega-\omega_0)^2 + \left(\frac{\gamma}{2}\right)^2}$

